		answer
7(a)	It is impossible to determine <u>exactly which nucleus</u> and <u>exactly when</u> a particular nucleus will disintegrate, although statistically, it is possible to predict the fraction of nuclei of a pure radioactive element that will decay after a certain length of time if its decay characteristics are known. *<u>Common mistakes:</u>* *Some students gave explanation for 'spontaneous' rather than 'random'. Must distinguish between the two.*	[1] for which nucleus [1] for when
7(b)	The half-life of a radioactive nuclide is the average time taken for a sample of the nuclide to decay to half its initial number of atoms/activity.	[1] for correct definition
7(c)(i)	${}^{131}_{53}\mathrm{I} \rightarrow {}^{0}_{-1}e + {}^{131}_{54}\mathrm{Xe}$	[1] for correct equation [1] for correct mass and atomic numbers of Xe

No.	Solution	Remarks
7(c)(ii)	At the start, Te-131 decays quickly to I-131 due to the short half-life as compared to the long half-life of decay of I-131. Hence, the activity decreases quickly as represented by PQ on the graph.	[1] for fast decay of Te-131 due to short half life [1] for quick decrease in activity
7(c)(iii)	$\text{gradient} = \dfrac{12.0 - 9.95}{0 - 3.0}$ $= -0.683$	[2] for correct gradient
7(c)(iv)1.	$A = A_0 e^{-\lambda t}$ $\lg A = \lg A_0 - (\lambda \lg e)t$ $\text{gradient} = -(\lambda \lg e)$ $= -0.683$ $\lambda = \dfrac{0.683}{\lg e}$ $= 1.57 \text{ hr}^{-1}$ $\text{half-life, } t_{1/2} = \dfrac{\ln 2}{\lambda}$ $= \dfrac{\ln 2}{1.57}$ $= 0.44 \text{ hours (shown)}$ *<u>Common mistakes:</u>* *Some students use the wrong linear equation for the initial part of the graph. They use ln instead of lg. That resulted in the wrong value for the decay constant and subsequently the half-life.*	[1] for correct equation of graph [1] for correct decay constant [1] for correct half-life
7(c)(iv)2.	From the graph, $A_0 = 1.0 \times 10^{12} \text{ Bq}$ $A_0 = \lambda N_0$ $N_0 = \dfrac{1.0 \times 10^{12}}{\frac{1.57}{60 \times 60}}$ $= 2.29 \times 10^{15}$ *<u>Common mistakes:</u>* *Almost all students did not change the decay constant to per second (or half-life to seconds) since the value of the initial activity is in Bq.*	[1] for correct expression and numerical substitution [1] for correct answer
7(c)(v)	The activity remains relatively constant after 6 hours. Almost all the Te must have decayed into I. The half-life of I is long, hence	[1] All Te decayed into I

No.	Solution	Remarks
	the activity remains constant for a long period of time as compared to Te. *Common mistakes:* *Many students thought that all the atoms have decayed to the stable Xe and there is no more activity which is wrong.*	[1] The activity remains constant due to long half life
7(c)(vi)	$\lg A = 9.325$ $A = 2.11 \times 10^{9}$ Bq Since all Te decayed to I, $N_0 = 2.29 \times 10^{15}$ $A = \lambda N_0$ $2.11 \times 10^{9} = \lambda \left(2.29 \times 10^{15}\right)$ $\lambda = 9.214 \times 10^{-7}\ \text{s}^{-1}$ $T_{1/2} = \dfrac{\ln 2}{9.214 \times 10^{-7}}$ $= 7.53 \times 10^{5}$ s $= 209$ hours *Common mistakes:* *Many students fail to realise that the initial number of Te nuclides is the same number of I nuclides at this point in time.*	[1] for calculating activity of I [1] for correct decay constant of I [1] for correct half-life
7(d)	The activities of the radioactive nuclides in this question have very high values hence the background radiation would be negligible.	[1] for correct explanation
7(e)	No, although the nuclides have high activities which is dangerous but their half-lives are relatively short so they will decay into stable isotopes quickly.	[1] for correct answer [1] for relatively short half-lives